

HeteroTac-Lift: Empirical Lift Admission via Heterogeneous Bilateral Tactile Servoing and Risk-Budgeted Compliance

Xingwei Chen, Shuaijun Liu, Feiyang You, Ningxin Su
Hong Kong University of Science and Technology (Guangzhou)

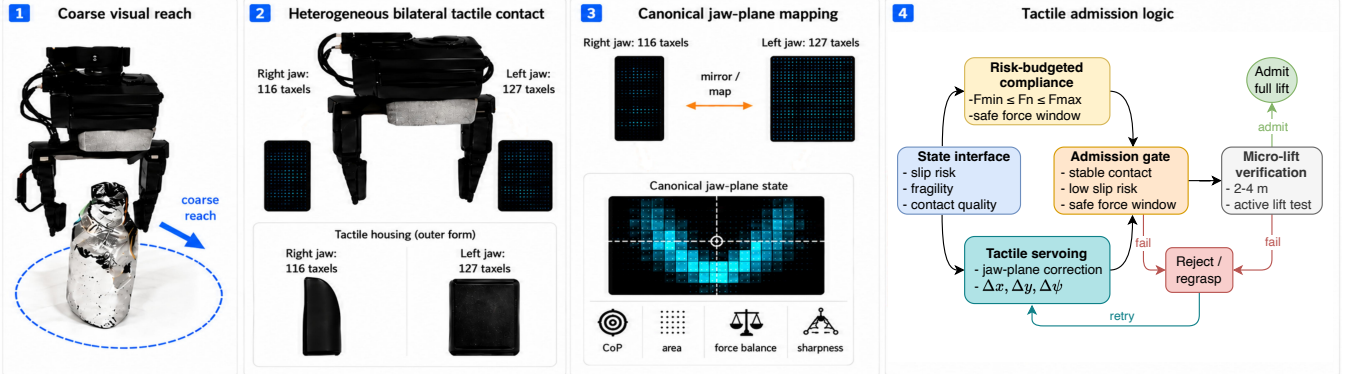


Fig. 1. System overview of HeteroTac-Lift. Vision provides only coarse near-field reach; from first contact, heterogeneous bilateral tactile sensing drives servoing, risk-budgeted compliance, and micro-lift verification for empirical lift admission (FAR).

Abstract—Vision can bring a gripper into a coarse near-field neighborhood, yet the final centimeters of contact-rich grasping remain brittle with transparent or reflective surfaces and narrow force margins for fragile or deformable objects. We present HeteroTac-Lift, a tactile-first framework that treats grasping as *empirical lift admission*: from first contact onward, tactile feedback governs whether the current grasp should be lifted, and admission quality is quantified by a measurable false accept rate (FAR) under a fixed lift primitive. Our system targets a practical setting with *heterogeneous* tactile arrays mounted on the two jaws (different taxel counts and noise characteristics) that output per-taxel 3D force vectors. HeteroTac-Lift combines: (i) a heterogeneous bilateral tactile *state interface* that aligns left/right sensing in a canonical jaw-plane via mirror alignment, mirror/swap consistency, and sensor-specific normalization, predicting continuous slip risk, fragility, and contact quality; (ii) a last-centimeter tactile servoing outer loop that refines wrist pose using a one-step local Jacobian update rather than multi-direction probing; and (iii) a risk-budgeted compliant inner loop that maintains grip force within dynamic bounds and resolves infeasible budgets through an unload-servo-reclose regrasp micro-cycle. Finally, a 2–4 mm *micro-lift* serves as an active verification action: apparently stable but unsafe grasps are rejected early and routed back to servoing before full lifting. Experiments on visually challenging objects and fragile/deformable items show improved lift success and reduced unsafe lift admissions (lower FAR), while preserving interpretability and reproducibility through explicit control structure and ablations.

I. INTRODUCTION

Reliable manipulation in real environments is often limited by the final centimeters before lift. In this regime, visual feedback can be unreliable due to occlusion (the gripper itself becomes the dominant occluder), specular/transparent surfaces, motion blur, and contact-induced camera motion. At the same time, fragile or low-friction objects impose

narrow force and safety envelopes: excessive squeeze causes damage, while insufficient normal force leads to slip and drop. These constraints make *lift* a safety-critical phase transition rather than a trivial continuation of reaching.

We therefore frame grasping as an *empirical lift-admission decision problem*. The central question is not only object localization, but whether the current contact state is ready for lift. This view shifts the paper’s focus from end success rate alone to decision quality and safety: false accepts (unsafe lifts), false rejects (unnecessarily refusing to lift), slip/drop events, over-force events, and table collisions. Importantly, we do not claim formal safety guarantees; instead, we provide calibrated risk signals and explicit, interpretable constraints that yield measurable reductions in unsafe admissions.

Our design principle is to use vision for coarse approach and tactile sensing for near-contact decisions. Vision is used only to bring the gripper into a coarse near-field neighborhood. Once contact begins, all phase transitions—contact refinement, force regulation, and lift permission—are governed by tactile evidence and proprioception. This separation is deliberate: even a strong visual model may degrade near contact, while tactile sensing provides local, high-rate evidence of contact distribution, centeredness, and incipient slip.

This paper targets a practical setting where the two gripper jaws are instrumented with *heterogeneous* tactile arrays (different taxel counts and noise characteristics) that output per-taxel 3D force vectors. We ask how to use such sensing to (i) refine coarse contact into a centered bilateral grasp, (ii) regulate grip forces under fragile/deformable constraints, and (iii) decide when lifting is permitted. The first-page system

overview in Fig. 1 summarizes the resulting perception–control–verification pipeline.

This paper makes three contributions:

- **Heterogeneous bilateral tactile state interface.** We introduce a reusable state and risk interface for heterogeneous left/right jaw sensing, aligning contact geometry in a canonical jaw-plane and enforcing mirror/swap consistency with sensor-specific normalization. The interface predicts continuous slip risk, fragility, and contact quality.
- **Last-centimeter tactile servoing with a one-step Jacobian update.** We define a reproducible tactile servoing error in the canonical jaw-plane and refine wrist pose using an online local Jacobian update, avoiding multi-direction probing and unbounded “search” behavior.
- **Risk-budgeted compliance with active lift verification.** We regulate grip forces within dynamic lower/upper bounds and resolve infeasible budgets via an unload–servo–reclose regrasp micro-cycle. A 2–4 mm micro-lift serves as an active verification action, and admission quality is quantified by a measurable false accept rate (FAR).

This work does not attempt to solve general open-world visual grasping. Instead, we target vision-constrained, contact-rich tabletop manipulation with fragile and slip-prone objects. Vision is used only for coarse reach initialization; it is not used as lift-admission truth. The technical contributions therefore lie in tactile state estimation, control, and admission logic *after* first contact.

II. RELATED WORK

Last-centimeter tactile servoing and refinement. A growing body of work treats contact as an opportunity for feedback-based refinement rather than a terminal condition. This includes robust visual–force servoing for insertion [1], purely tactile insertion strategies such as extremum-seeking “wiggling” [2], and tactile-reactive grasp adjustment at high bandwidth [3]. Recent tactile-only refinement methods further emphasize contact-driven correction when vision is unreliable [4], and shear-focused tactile servoing has been explored in sim-to-real settings [5]. This line of work motivates our tactile servoing outer loop: we target a practical two-finger gripper with *heterogeneous* jaw arrays, and use an explicit, reproducible jaw-plane servoing error with a one-step local Jacobian update to avoid multi-direction probing and unbounded “search”.

Visuo-tactile representations and policy learning. Learning-based visuomotor policies, including diffusion-style imitation learning [6], [7] and recent slow-fast visuo-tactile variants [8], [9], have shown strong performance in contact-rich tasks, while representation learning increasingly aims for transfer across sensors and tasks [10], [11], [12], [13]. In contrast to end-to-end action generation, our learning component is intentionally limited to a heterogeneous bi-tactile *state interface*: it predicts calibrated risk and contact states (slip, fragility, contact quality) that parameterize an interpretable controller and lift-admission decision.

Slip, shear, and fragile-object handling. Slip detection and shear-aware feedback are central for reliable grasping under low friction and narrow force margins [14], [15], [16]. Complementary tactile-reactive control frameworks also highlight explicit adaptation under uncertain contact mechanics [17], [18]. HeteroTac-Lift operationalizes these signals as a dynamic force budget with an explicit conflict handler: when preventing slip would require unsafe squeezing, we trigger an unload–servo–reclose regrasp micro-cycle rather than increasing grip force.

Constraints, safety filtering, and verification. Safety-aware manipulation has been studied via constraint layers and safe filtering in contact-rich grasping [19], and tactile sensing has also been used to improve state estimation under partial observability (e.g., pose/contact estimation from low-resolution tactile arrays) [20], [21]. We take a complementary decision-centric view: we do not claim formal guarantees, but quantify *empirical lift admission* via a false accept rate (FAR) and introduce a 2–4 mm micro-lift as an active verification action. To support threshold-based decisions, we adopt risk calibration methods to align predicted risks with empirical event frequencies [22], [23].

III. METHODOLOGY

We assume an upstream vision module provides a coarse *near-field* approach pose, bringing the gripper close to the target object but not necessarily aligned. From the first contact onward, all decisions—contact refinement, force regulation, and whether lifting is permitted—are governed by a tactile closed loop. We call the resulting manipulation objective *empirical lift admission*: a grasp is successful only if it enters a stable, risk-budgeted *bilateral* contact regime and passes an active *micro-lift* verification before executing a full lift.

Our goal is not merely to “close the fingers,” but to achieve a *verifiably liftable* grasp under a narrow force window (e.g., fragile/deformable items) and poor visual observability (e.g., transparent/reflective items). We define *certification* in an empirical sense: under a fixed distribution of objects and a fixed lift primitive, we measure the *false accept rate* (FAR)—the probability that the system admits lifting and then encounters a safety event within a prescribed time window. We do *not* claim formal safety guarantees; instead we provide calibrated risk signals and explicit, interpretable constraints that enable measurable reductions in FAR.

A. Heterogeneous Tactile Geometry and State Interface

Our hardware consists of a two-finger gripper equipped with heterogeneous Paxini tactile arrays mounted on the inner surfaces of the left and right jaws. At time t , the left M2324 array outputs $\mathbf{M}_L(t) \in \mathbb{R}^{N_L \times 3}$ with $N_L = 127$ taxels, and the right S2716 array outputs $\mathbf{M}_R(t) \in \mathbb{R}^{N_R \times 3}$ with $N_R = 116$ taxels. Each taxel provides a 3D force vector $\mathbf{f}_i = (f_{x,i}, f_{y,i}, f_{z,i})$. We assume known taxel locations $\mathbf{p}_i$ in a jaw-local parameterization obtained from a gripper-mounted YAML/CAD specification. Fig. 2 shows

the gripper-mounted sensor placement and the heterogeneous left/right taxel layouts used by the controller.

To make left–right geometric comparisons well-defined (e.g., CoP differences), we introduce a *canonical jaw-plane* 2D coordinate frame that parameterizes the inner surface of each jaw. This requires only a known rigid mirror alignment from the gripper assembly. Specifically, we apply a fixed planar mirror transform $\mathbf{R}_{\text{mirror}}$ (optionally with a constant translation) to map right-jaw coordinates into the left-jaw canonical frame:

$$\tilde{\mathbf{p}}_{R,i} = \mathbf{R}_{\text{mirror}} \mathbf{p}_{R,i}, \quad \tilde{\mathbf{c}}_R = \mathbf{R}_{\text{mirror}} \mathbf{c}_R. \quad (1)$$

All left–right geometric errors are computed in this shared jaw-plane parameterization (e.g., $\mathbf{c}_L - \tilde{\mathbf{c}}_R$), avoiding ill-posed subtraction across unrelated local frames.

Minimal axis/scale calibration. Directly treating f_z as the jaw-normal force is brittle under mounting tolerances (adhesive layers, slight bending, axis misalignment). Instead, for each side $S \in \{L, R\}$ we define an assembly-provided approximate normal unit vector $\mathbf{n}_S$ and fit a minimal scalar correction s_S (sign and scale) using a one-direction press test (e.g., loading with a flat plate). We then decompose each taxel force into normal and tangential magnitudes:

$$\begin{aligned} f_{n,i}^S &= \max(\mathbf{n}_S^\top (s_S \mathbf{f}_i^S), 0), \\ f_{t,i}^S &= \|\mathbf{s}_S \mathbf{f}_i^S - (\mathbf{n}_S^\top (s_S \mathbf{f}_i^S)) \mathbf{n}_S\|. \end{aligned} \quad (2)$$

This protocol is intentionally minimal and reproducible; it prevents “axis-assumption” failure modes without introducing a heavy calibration burden.

Top- k normalization. Heterogeneous arrays exhibit different noise spectra and taxel counts; naive thresholding can bias contact area and shape statistics and undermine cross-sensor alignment. We therefore *fix* a Top- k scheme across all experiments. For each side S , we select $k_S = \rho N_S$ taxels with largest $\|\mathbf{f}_i^S\|$ (constant ratio ρ), and compute all statistics on this subset. To preserve a meaningful “area” proxy under Top- k , we use the *effective support size*

$$A_{\text{eff}}^S = \frac{(\sum_i w_i^S)^2}{\sum_i (w_i^S)^2 + \varepsilon}, \quad (3)$$

which approaches 1 for point-like contact and k_S for broadly distributed contact.

Controllable tactile state. For each side $S \in \{L, R\}$, using the Top- k active set and jaw-plane locations, we compute interpretable, control-friendly statistics:

$$F_n^S = \sum_i f_{n,i}^S, \quad F_t^S = \sum_i f_{t,i}^S, \quad (4)$$

$$\mathbf{c}^S = \frac{\sum_i \mathbf{p}_i^S w_i^S}{\sum_i w_i^S + \varepsilon}, \quad w_i^S \in \{f_{n,i}^S, \|\mathbf{f}_i^S\|\}, \quad (5)$$

$$\Sigma^S = \sum_i w_i^S (\mathbf{p}_i^S - \mathbf{c}^S)(\mathbf{p}_i^S - \mathbf{c}^S)^\top, \quad (6)$$

$$\text{peak}^S = \max_i f_{n,i}^S, \quad \text{sharp}^S = \frac{\text{peak}^S}{F_n^S + \varepsilon}. \quad (7)$$

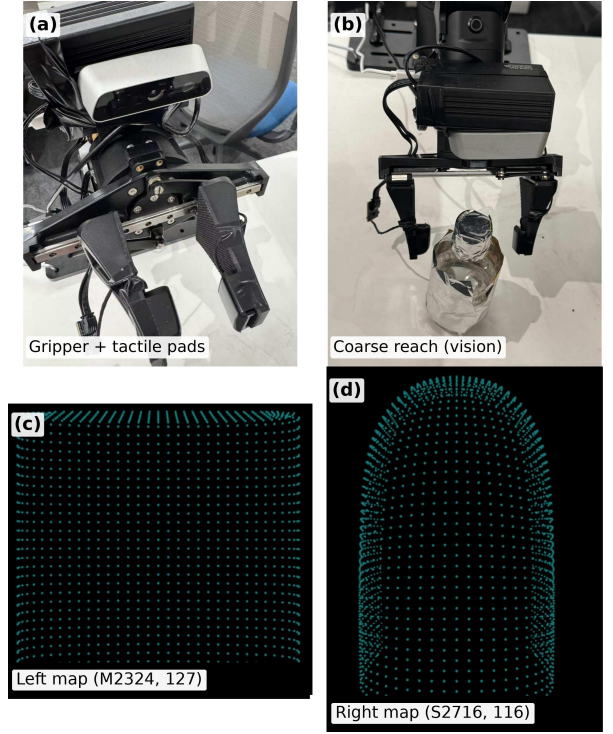


Fig. 2. Hardware and tactile geometry for heterogeneous jaw sensing. The figure shows the gripper-mounted sensor placement and the heterogeneous left/right taxel layouts (M2324: 127 taxels; S2716: 116 taxels) used by the method’s canonical jaw-plane representation and state interface.

We mirror the right-jaw CoP via Eq. (1) to form $\tilde{\mathbf{c}}_R$. We then define a *tactile servoing error* that is fixed and reproducible:

$$\mathbf{e}(t) = \begin{bmatrix} F_n^L - F_n^R \\ A_{\text{eff}}^L - A_{\text{eff}}^R \\ \mathbf{c}_L - \tilde{\mathbf{c}}_R \\ \phi(\Sigma^L, \Sigma^R, \text{sharp}^L, \text{sharp}^R) \end{bmatrix}, \quad (8)$$

where $\phi(\cdot)$ collects a small number of shape/contact-quality indicators derived from second moments and sharpness (e.g., point contact vs. edge contact vs. surface contact, and incipient torsional trends). These quantities are computed over a short temporal window (0.5–1.0 s) to support both learning-based state estimation and controller gating.

Learned bi-tactile state interface. We learn *state* and *risk* signals (not actions), providing a reusable interface for control:

$$\begin{aligned} \hat{p}_{\text{slip}}(t) &\in [0, 1], \\ \hat{c}_{\text{fragile}}(t) &\in \mathbb{R}, \\ \hat{q}_{\text{contact}}(t) &\in \mathbb{R} \text{ or } \{1, \dots, C\}. \end{aligned} \quad (9)$$

Here $\hat{p}_{\text{slip}}$ is a continuous short-horizon slip/instability risk, $\hat{c}_{\text{fragile}}$ quantifies deformability/fragility, and $\hat{q}_{\text{contact}}$ scores contact quality (e.g., single-sided, point, edge, torsional).

Architecture. Each side uses an independent encoder Enc_S operating on the point set $\{(\mathbf{p}_i^S, \mathbf{f}_i^S)\}$ to produce latent embeddings $\mathbf{z}_L(t), \mathbf{z}_R(t)$. We fuse them with an explicitly symmetric operator:

$$\mathbf{z}(t) = \text{MLP}([\mathbf{z}_L, \mathbf{z}_R, \mathbf{z}_L - \mathbf{z}_R, \mathbf{z}_L \odot \mathbf{z}_R]), \quad (10)$$

followed by a single-layer causal GRU over the 0.5–1.0 s window, and three output heads for Eq. (9).

Heterogeneous alignment. To turn left–right heterogeneity into a reusable method component, we enforce *mirror/swap consistency*: when we mirror and swap left/right tactile observations, the predicted state should remain consistent in the canonical jaw-plane. In addition, we include a small sensor-specific affine normalization layer per side to absorb scale and noise differences. Finally, we calibrate only $\hat{p}_{\text{slip}}$ (e.g., temperature scaling or isotonic regression) so that thresholds used in gating and force budgeting have consistent semantics [22], [23].

B. Tactile Servoing and Risk-Budgeted Compliance

Our controller consists of (i) a fast inner loop that regulates grasp forces within a risk-budgeted feasible band, and (ii) a slower outer loop that performs last-centimeter tactile servoing to converge from coarse contact to a stable bilateral contact basin.

Inner Loop: Risk-Budgeted Compliant Grasping. The inner loop runs as a fast reflex and enforces a feasible interval for the total normal force:

$$\begin{aligned} F_{\min}(t) &= h(\hat{p}_{\text{slip}}(t), \text{payload_proxy}), \\ F_{\max}(t) &= g(\hat{c}_{\text{fragile}}(t), \text{peak}(t), A_{\text{eff}}(t)), \end{aligned} \quad (11)$$

where $F_{\min}$ prevents slip and $F_{\max}$ prevents damage (using fragility and peak/sharpness cues). We command gripper velocity/force (depending on hardware capability) with a PI controller and saturation such that $F_n(t)$ is maintained within $[F_{\min}(t), F_{\max}(t)]$.

If the interval becomes infeasible ($F_{\min}(t) > F_{\max}(t)$), we forbid further force increases and trigger the regrasp micro-cycle described below. This is crucial for fragile objects where increasing force to prevent slip can immediately cause deformation or crushing.

Outer Loop: Last-Centimeter Tactile Servoing. The outer loop runs at a slower rate and refines wrist pose with small bounded primitives to drive $\mathbf{e}(t)$ (Eq. (8)) into a stable, centered bilateral contact region. We optimize over a low-dimensional action $\Delta \mathbf{u} = [\Delta x, \Delta y, \Delta \psi]^\top$ (millimeter- and degree-scale), optionally coupled with a small open–close micro action. We enforce a strict per-step clip and a total budget of K steps to avoid “search” behavior.

Instead of probing multiple directions per step, we maintain an online local linear model

$$\Delta \mathbf{e} \approx \mathbf{J} \Delta \mathbf{u}, \quad (12)$$

and fit $\mathbf{J}$ via a sliding-window least squares or RLS update using recent $(\Delta \mathbf{u}, \Delta \mathbf{e})$ pairs. We then apply a one-step descent update:

$$\Delta \mathbf{u} = \text{clip}(-\eta \mathbf{J}^+ \mathbf{e}(t)), \quad (13)$$

where $\mathbf{J}^+$ is the pseudoinverse and η is a small step size.

When tangential force or sharpness is high, we *unload* to a safe light-contact level before executing $\Delta \mathbf{u}$, reducing the risk of dragging, scratching, or pushing objects.

Regrasp micro-cycle. When the force budget becomes infeasible ($F_{\min} > F_{\max}$) or contact quality is poor (low $\hat{q}_{\text{contact}}$, indicating point/edge/single-sided contact), we execute a fixed three-stage primitive:

$$\begin{aligned} \text{unload: } & F_n \rightarrow F_{\text{reset}}, \\ \text{servo: } & \text{one-step } \Delta \mathbf{u}, \\ \text{re-close: } & F_n \rightarrow [F_{\min}, F_{\max}]. \end{aligned} \quad (14)$$

During *unload*, we restrict wrist motion to be minimal to prevent frictional dragging on deformable or easily scratched objects. This micro-cycle is the key mechanism for fragile items: it improves contact geometry instead of “forcing through” by increasing grip force.

C. Lift Admission and Micro-Lift Active Verification

Admission gate. We permit lifting only when all hard guards pass (bilateral contact, peak/sharpness below limits, and $\mathbf{e}(t)$ stable for T ms) and the learned-state thresholds are satisfied:

$$\hat{p}_{\text{slip}}(t) < \alpha, \quad \hat{q}_{\text{contact}}(t) > \gamma. \quad (15)$$

Micro-lift as active test. Upon admission, we perform a short micro-lift of 2–4 mm and observe the tactile state transition. This local range is large enough to expose early slip or contact imbalance, but small enough to abort before a full drop or collision. If any of the following worsen beyond thresholds,

$$\Delta \hat{p}_{\text{slip}}, \quad \Delta \|\mathbf{c}_L - \tilde{\mathbf{c}}_R\|, \quad \Delta A_{\text{eff}}, \quad (16)$$

we reject and return to the regrasp micro-cycle or tactile servoing; otherwise we execute a soft-start full lift. This turns lifting into a verifiable phase transition under poor vision and contact uncertainty.

Empirical certification metric. Let $E_{\Delta T}$ denote the event that a safety event occurs within a fixed window ΔT after lift admission. We define the false accept rate as

$$\text{FAR} = \Pr(E_{\Delta T} \mid \text{admit}), \quad (17)$$

where “safety event” includes post-admission slip/drop, excessive peak/sharpness, or post-admission table contact during a fixed lift primitive and a fixed evaluation window ΔT . FAR directly quantifies the empirical certification quality of lift admission.

Overall, our method is deliberately structured to avoid “kitchen-sink” complexity, with three top-level contributions: (i) a heterogeneous bi-tactile *state interface* enabled by mirror/swap consistency and sensor-specific normalization; (ii) last-centimeter tactile servoing defined in a canonical jaw-plane and implemented via a one-step local Jacobian update; and (iii) risk-budgeted compliant grasping with a regrasp micro-cycle and micro-lift active verification, yielding measurable reductions in FAR under challenging visual and fragility regimes.

TABLE I
PROTOCOL PARAMETERS USED IN ALL EXPERIMENTS.

Parameter	Value
Top- k ratio ρ	0.15
Outer-loop step cap K	6
Outer-loop step size η	0.30
Jacobian fit window W (steps)	4
Micro-lift displacement d_μ	3 mm
Admission stability window T	120 ms
FAR window ΔT	200 ms
Slip-risk threshold α	0.30
Contact-quality threshold γ	0.70
Unloading target F_{reset}	$0.35 F_{\text{max}}$
Peak/sharpness limits	0.80 N / 0.55

IV. EXPERIMENTS

A. Experimental Setup

The paper targets a deployment regime where the final centimeters before lift dominate safety and reliability. Our evaluation therefore emphasizes *lift-admission quality* (decision correctness) alongside end-to-end task success. Concretely, we test three hypotheses:

- **H1: Last-centimeter convergence** Tactile servoing in the canonical jaw-plane increases the conversion rate from coarse reach to stable bilateral contact and lift success, especially under visually challenging conditions.
- **H2: Fragility compliance** Risk-budgeted compliance with regrasp micro-cycles reduces unsafe events (overforce / excessive peak pressure and slip/drop) compared to fixed-force or slip-only heuristics.
- **H3: Empirical admission** Micro-lift active verification reduces unsafe lift admissions (lower FAR) under matched action and time budgets.

We evaluate on a real 6-DoF AgileX Piper arm with a parallel gripper. The inner surfaces of the left and right jaws are instrumented with heterogeneous Paxini M2324 ($N_L = 127$) and S2716 ($N_R = 116$) tactile arrays, respectively, each returning a 3D force vector. An eye-in-hand RGB camera provides a first-person view for coarse initialization. The tactile closed loop executes bounded primitives (small wrist refinements, controlled closure adjustments, and micro-lift).

The tactile servo loop runs at 50 Hz. The learned state interface operates on 0.5 s windows (25 samples): each jaw uses a two-layer PointNet-style MLP encoder (64-d latent), the fused representation is processed by a single-layer GRU (hidden size 128), and three MLP heads predict $\hat{p}_{\text{slip}}$, $\hat{c}_{\text{fragile}}$, and $\hat{q}_{\text{contact}}$. Training uses Adam (10^{-3}), binary cross-entropy for slip, smooth- L_1 for fragility, and cross-entropy for contact quality; only the slip-risk head is post-hoc calibrated by temperature scaling on the validation split. The outer-loop wrist refinement is capped to millimeter- and degree-scale updates in Δx , Δy , $\Delta \psi$ and limited to K steps per trial; the micro-lift uses a fixed 2–4 mm displacement and verification window ΔT . Table I lists the fixed parameters.

Fig. 3 illustrates how the outer-loop tactile servoing (error reduction), inner-loop risk-budgeted force regulation, the

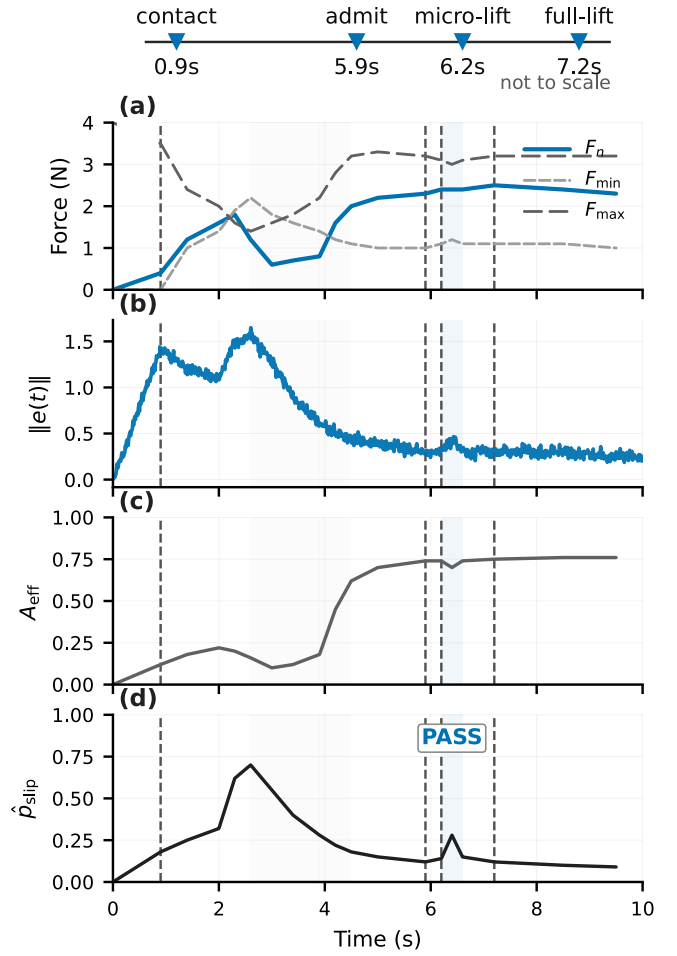


Fig. 3. Representative trial trace showing risk-budgeted compliance and tactile servoing over time, including admission and micro-lift verification.

regrasp micro-cycle, and micro-lift verification play out on a single trial.

B. Evaluation Protocol

Objects and trials. We study tabletop grasp-and-lift trials with objects that stress vision and contact safety:

- **Vision-constrained:** near-contact occlusion by the gripper, specular/transparent surfaces, and motion blur.
- **Fragile / deformable:** objects that are easily damaged by high peak pressure or deform under small force increases.
- **Low-friction / slip-prone:** smooth objects and small contact patches that generate ambiguous early contact signatures.

We group evaluation objects into two sets aligned with the paper’s motivation. Table II lists the objects used in this paper.

Each trial follows the same funnel:

$$\text{reach} \rightarrow \text{bilateral} \rightarrow \text{admit} \rightarrow \text{micro_pass} \rightarrow \text{lift_success/fail.} \quad (18)$$

Target detection is outside the scope of this work. We therefore use an off-the-shelf vision routine to produce a

TABLE II
OBJECT SETS USED IN EVALUATION.

Set	Examples
Vision-constrained	Transparent plastic bottle; plastic bottle wrapped with reflective foil
Fragile/deformable	Paper cup; crunchy noodle block

coarse 2D target neighborhood and move the gripper to a nearby *taught approach node*. This “active vision + taught-node” initialization is a standard engineering practice to avoid dependence on precise depth at close range. Importantly, all contact-phase transitions and lift admission are decided from tactile evidence; coarse reach only determines whether the system starts the tactile funnel near the object.

Data and splits. Real data collection logs synchronized tactile signals, joint states, gripper commands, and an outcome label with failure taxonomy (drop, over_force, table_contact, push_away, budget_exhausted, no_contact). For the main evaluation, we executed 25 matched trials for each object-method pair, yielding 100 trials per method across the four objects and 500 real-robot evaluation trials in total over B0–B4. Because each method has 100 trials, success, admission, damage, and failure-mode percentages in Fig. 4 correspond directly to counts out of 100, whereas FAR is normalized by the number of admitted lifts. Each object-method pair uses the same taught approach node, object pose template, action caps, failure taxonomy, and admission criteria. Pre-lift failures such as no_contact, push_away, and budget_exhausted remain in the denominator, and fragile/deformable objects are reset and visually checked between trials to avoid mixing nominal and visibly degraded contacts.

For the learned tactile state interface, we extract 2,000 contact-phase windows from real logs and split them by collection session into 1,200 training, 400 validation, and 400 held-out test windows. Calibration is fitted only on the validation split, and all reported controller thresholds remain frozen on the held-out test and deployment runs.

Baselines. We compare tactile lift-admission designs under matched budgets:

- **B0: Vision-only** Reach $\rightarrow$ close $\rightarrow$ lift with fixed grip parameters (no tactile closed-loop refinement).
- **B1: No tactile servoing** Inner-loop force regulation and admission are enabled, but wrist pose is not refined (outer-loop disabled).
- **B2: No risk budget** Tactile servoing is enabled, but grip force uses fixed bounds without fragility-aware budgeting and without regrasp micro-cycles.
- **B3: No micro-lift** Full HeteroTac-Lift but skipping the micro-lift verification (admit directly to full lift).
- **B4: Full** HeteroTac-Lift with heterogeneous state interface, tactile servoing, risk-budgeted compliance, and micro-lift verification (ours).

Metrics. We report both *outcome* and *decision* metrics:

- **Outcome:** lift success rate; drop/slip rate; over-

force/peak-pressure events; table-contact and push-away rates.

- **Admission quality:** false accept rate (FAR) for lift admission, defined as “lift was allowed” and a safety-critical outcome occurred during the subsequent lift primitive. In our trials, FAR-critical events are post-admission slip/drop, over-force/peak-pressure, and post-admission table contact; push-away and no-contact are reported separately as pre-lift failure modes. We also report the *abort rate* (no lift attempt within budget) to characterize conservatism.
- **Process:** funnel conversion rates for (18) and per-stage latency.
- **Efficiency:** attempts per trial and mean completion time.

Budget matching. Budget fairness is essential because tactile servoing and micro-lifts introduce additional actions. All methods therefore use matched caps on:

- number of alignment nudges and closure adjustments,
- maximum micro-lift attempts,
- maximum per-trial wall-clock time.

If a method introduces an additional probe (micro-lift), baselines receive equivalent allowance (either additional alignment budget or additional time), and we report both raw and budget-normalized results.

To quantify how much coarse reach accuracy is required, we evaluate robustness to initialization error by perturbing the taught approach node by controlled offsets (in joint space or end-effector plane) and measuring conversion in (18). This isolates the regime where touch-first admission remains effective without requiring continuous visual servoing. We report success and FAR as a function of the perturbation magnitude.

C. Main Results and Ablations

Fig. 4 summarizes the matched-budget B0–B4 ablations in terms of lift success, admission behavior, FAR, damage on fragile/deformable objects, and failure modes.

Panels (a), (c), and (e) of Fig. 4 report success, FAR, and failure-mode counts under the same matched budgets. B4 achieves 76/100 lift success; FAR is 5/86 (5.8%), lower than B3 (19.3%), B2 (28.1%), B1 (26.5%), and B0 (47.0%). The gap between B3 and B4 indicates that the micro-lift is not merely an additional motion primitive, but an admission filter that rejects unstable grasps before committing to a full lift.

D. Qualitative Frame Strips

Fig. 5 shows representative frame strips for all four evaluation objects, each following the same sequence: pre-contact, first contact, servo/regrasp, admission and micro-lift, and full lift.

V. LIMITATIONS

Our current study has four main limitations.

First, the deployment focus is tabletop contact-rich manipulation with bounded object categories. Generalization to

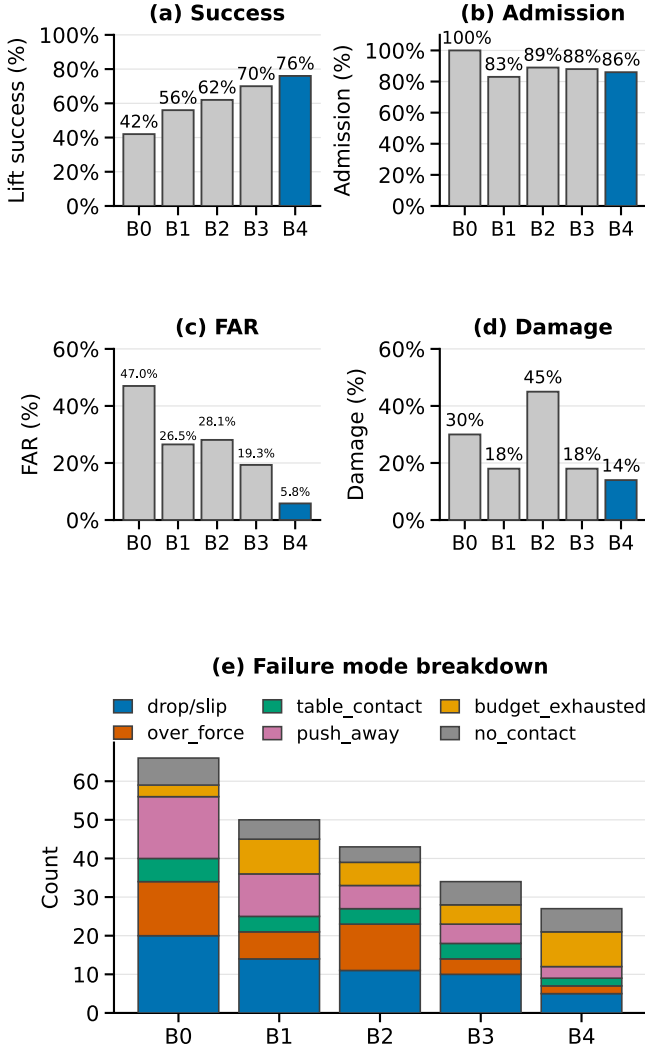


Fig. 4. Main matched-budget results and ablations for HeteroTac-Lift. B0–B4 are the ablation variants: (a) lift success, (b) admission rate, (c) FAR (unsafe/admitted), (d) damage on fragile/deformable objects, and (e) failure-mode counts, including budget exhaustion.

highly dynamic scenes, severe external perturbations, and fully unstructured clutter still needs dedicated validation.

Second, although vision is intentionally restricted to coarse localization, initialization error can still increase recovery burden. This is a design choice for robustness near contact, but it may reduce efficiency in settings where vision is reliable. In our implementation, coarse reach is supported by a small set of taught approach nodes, which trades generality for deployment stability.

Third, the risk thresholds and calibration quality depend on the representativeness of the collected real data. Distribution shift across hardware wear, sensor aging, adhesive creep, or unseen surface materials can degrade admission quality. Similarly, our minimal force-axis calibration (sign/scale along an approximate normal) is designed to be lightweight rather than perfect.

Fourth, our control design favors controllability and interpretability over maximal policy expressiveness. While

we avoid multi-direction probing by using a one-step local Jacobian update, this outer-loop approximation can be sensitive when contact mechanics change abruptly (e.g., rolling contacts). End-to-end policies might achieve better peak performance in specific narrow regimes, but typically with weaker safety semantics.

These limitations motivate future work on adaptive calibration under drift, broader object distributions, and tighter integration between coarse vision uncertainty and tactile admission thresholds. They also define concrete evaluation axes for future deployments: object diversity, sensor drift, support-surface variation, and perturbation robustness. Treating these axes as separate stress tests would help distinguish admission failures caused by sensing uncertainty from those caused by contact geometry or force-budget infeasibility.

At the same time, this work intentionally prioritizes controllability and deployability over maximal model capacity. The resulting system is therefore best interpreted as a tactile decision-and-control framework for near-contact lifting, rather than a general end-to-end solution for arbitrary visual manipulation.

VI. CONCLUSION

We presented HeteroTac-Lift, a tactile-first framework for vision-limited contact-rich manipulation, centered on empirical lift admission rather than pure grasp execution. Vision is used only for coarse near-field reach initialization; from first contact onward, heterogeneous bilateral tactile sensing governs contact refinement, force regulation, and whether lifting is permitted.

Our method combines a heterogeneous bilateral tactile state interface, a last-centimeter tactile servoing outer loop with a one-step Jacobian update, and a risk-budgeted compliant inner loop with unload–servo–reclose regrasp micro-cycles. A 2–4 mm micro-lift serves as an active verification action, and admission quality is quantified by the false accept rate (FAR) under a fixed lift primitive. Together, these components provide deployable safety semantics: explicit pass/reject logic, bounded recovery, and interpretable failure modes, supporting reliable lifting when visual reliability drops near contact and when force margins are narrow.

Beyond the specific gripper and object set used here, the framework suggests a practical way to reuse asymmetric tactile hardware without requiring identical sensing modules on both jaws. By expressing contact evidence, servoing errors, force limits, and lift admission in a common jaw-plane interface, the same decision logic can be audited and retuned as sensors, fingertips, or object distributions change.

Future work will focus on larger-scale real data, adaptive calibration under long-term drift, and stronger uncertainty-aware coupling between coarse visual initialization and tactile admission. In practice, this turns near-contact grasping into an interpretable pass/reject decision process rather than a purely reactive closure policy.

REFERENCES

- [1] H. Chang, A. Boularias, and S. Jain, “Insert-One: One-shot robust visual-force servoing for novel object insertion with 6-DoF tracking,”

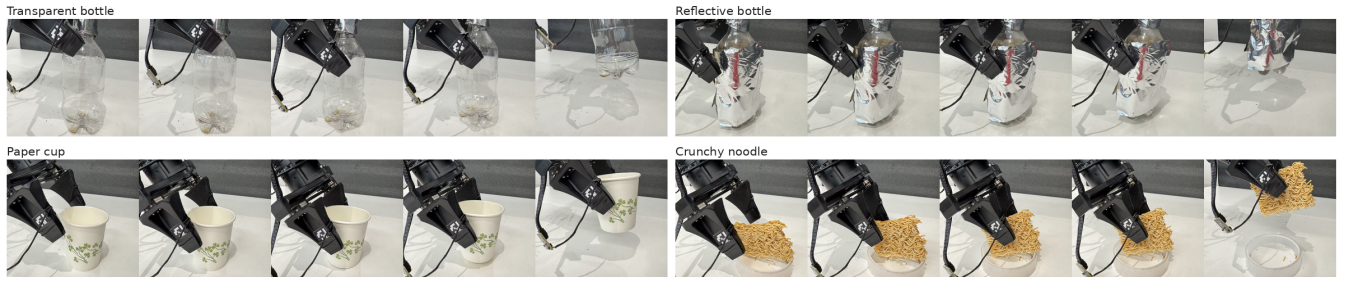


Fig. 5. Representative frame strips for four objects (two vision-constrained and two fragile/deformable). In each strip, frames proceed left-to-right: pre-contact, first contact, servo/regrasp, admit+micro-lift, and full lift.

- in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, Oct. 2024, pp. 2935–2942, doi: 10.1109/IROS58592.2024.10801884.
- [2] L. Burner, P. Mantripragada, G. M. Caddeo, L. Natale, C. Fermüller, and Y. Aloimonos, “Extremum seeking controlled wiggling for tactile insertion,” *arXiv preprint arXiv:2410.02595*, 2024.
 - [3] Y. Lee, T.-Y. Lin, A. Alexiev, and S. Kim, “High-bandwidth tactile-reactive control for grasp adjustment,” *arXiv preprint arXiv:2509.15876*, 2025.
 - [4] S. Wang, H. Zhou, D. Xiang, and Y. You, “TacRefineNet: Tactile-only grasp refinement between arbitrary in-hand object poses,” *arXiv preprint arXiv:2509.25746*, 2025.
 - [5] K. M. Freud, Y. Lin, and N. F. Lepora, “SimShear: Sim-to-Real shear-based tactile servoing,” *arXiv preprint arXiv:2508.20561*, 2025.
 - [6] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, “Diffusion policy: Visuomotor policy learning via action diffusion,” in *Robotics: Science and Systems XIX*, Jul. 2023, doi: 10.15607/RSS.2023.XIX.026.
 - [7] Y. Ze, G. Zhang, K. Zhang, C. Hu, M. Wang, and H. Xu, “3D diffusion policy: Generalizable visuomotor policy learning via simple 3D representations,” in *Robotics: Science and Systems XX*, Jul. 2024, doi: 10.15607/RSS.2024.XX.067.
 - [8] H. Xue, J. Ren, W. Chen, G. Zhang, F. Yuan, G. Gu, H. Xu, and C. Lu, “Reactive diffusion policy: Slow-fast visual-tactile policy learning for contact-rich manipulation,” in *Robotics: Science and Systems XXI*, 2025, doi: 10.15607/RSS.2025.XXI.052.
 - [9] Y. Wu, Z. Chen, F. Wu, L. Chen, L. Zhang, Z. Bing, A. Swikir, S. Haddadin, and A. Knoll, “TacDiffusion: Force-domain diffusion policy for precise tactile manipulation,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2025, pp. 11 831–11 837, doi: 10.1109/ICRA55743.2025.11127334.
 - [10] J. Zhao, Y. Ma, L. Wang, and E. H. Adelson, “Transferable tactile transformers for representation learning across diverse sensors and tasks,” in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270, 2025, pp. 3766–3779.
 - [11] C. Higuera, A. Sharma, C. K. Bodduluri, T. Fan, P. Lancaster, M. Kalakrishnan, M. Kaess, B. Boots, M. Lambeta, T. Wu, and M. Mukadam, “Sparsh: Self-supervised touch representations for vision-based tactile sensing,” in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270, 2025, pp. 885–915.
 - [12] V. Pattabiraman, Y. Cao, S. Haldar, L. Pinto, and R. Bhirangi, “Learning precise, contact-rich manipulation through uncalibrated tactile skins,” in *ICRA 2025 Workshop on Beyond Pick and Place*, 2025, openReview workshop paper.
 - [13] L. Heng, H. Geng, K. Zhang, P. Abbeel, and J. Malik, “ViTacFormer: Learning cross-modal representation for visuo-tactile dexterous manipulation,” *arXiv preprint arXiv:2506.15953*, 2025.
 - [14] N. Jawale, N. Kaur, A. Santoso, X. Hu, and X. Chen, “Learned slip-detection-severity framework using tactile deformation field feedback for robotic manipulation,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, Oct. 2024, doi: 10.1109/IROS58592.2024.10802687.
 - [15] C. J. Ford, H. Li, M. G. Catalano, M. Bianchi, E. Psomopoulou, and N. F. Lepora, “Shear-based grasp control for multi-fingered underactuated tactile robotic hands,” *IEEE Transactions on Robotics*, 2025, doi: 10.1109/TRO.2025.3563046.
 - [16] S. Lyu, L. Sheng, F. Wang, W. Zhang, W. Lin, Z. Jia, D. Navarro-Alarcon, and G. Guo, “Phy-Tac: Toward human-like grasping via physics-conditioned tactile goals,” *arXiv preprint arXiv:2511.01520*, 2025.
 - [17] Z. Xu and Y. She, “LeTac-MPC: Learning model predictive control for tactile-reactive grasping,” *IEEE Transactions on Robotics*, vol. 40, pp. 4376–4395, 2024, doi: 10.1109/TRO.2024.3463470.
 - [18] Z. Zhao, Y. Li, W. Li, Z. Qi, L. Ruan, Y. Zhu, and K. Althoefer, “Tac-Man: Tactile-informed prior-free manipulation of articulated objects,” *IEEE Transactions on Robotics*, vol. 41, pp. 538–557, 2025, doi: 10.1109/TRO.2024.3508134.
 - [19] Y. Kim, J. Kim, A. H. Li, A. D. Ames, and C. Sloth, “Robust adaptive safe robotic grasping with tactile sensing,” in *Proceedings of the European Control Conference (ECC)*, 2025, pp. 2531–2538, doi: 10.23919/ECC65951.2025.11186832.
 - [20] L. Mack, F. Grüniger, B. A. Richardson, R. Lendway, K. J. Kuchenbecker, and J. Stueckler, “Visuo-Tactile object pose estimation for a multi-finger robot hand with low-resolution in-hand tactile sensing,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, May 2025, pp. 12 401–12 407, doi: 10.1109/ICRA55743.2025.11127966.
 - [21] M. Van der Merwe, K. Ota, D. Berenson, N. Fazeli, and D. K. Jha, “Simultaneous extrinsic contact and in-hand pose estimation via distributed tactile sensing,” *arXiv preprint arXiv:2512.23856*, 2025.
 - [22] L. Tao, M. Dong, D. Liu, C. Sun, and C. Xu, “Calibrating a deep neural network with its predecessors,” in *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*, Aug. 2023, pp. 4271–4279, doi: 10.24963/IJCAI.2023/475.
 - [23] C. Marx, S. Zalouk, and S. Ermon, “Calibration by distribution matching: Trainable kernel calibration metrics,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.